

Suspicious Transaction Investigation Summary

Portfolio Case Study — Fraud & AML Detection Dashboard, Maple Ridge Financial Group (fictional)

This is a demonstration case study built for a portfolio project. All account identifiers, names, and transactions are drawn from the IBM AML synthetic transaction dataset (reframed in CAD) and community-detection output from this project's pipeline; no real financial institution, customer, or account is involved. The structure below mirrors the field organization of FINTRAC's Suspicious Transaction Report (STR) guidance for illustrative purposes only and is not a filed regulatory report.

1. Report Metadata

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Case reference: CASE-2026-0142 (illustrative)

Detection source: Fraud & AML Detection Dashboard — anomaly scoring (Isolation Forest / XGBoost) + network community detection (Louvain)

Reporting entity: Maple Ridge Financial Group (fictional, for portfolio purposes)

Report status: Draft — recommended for STR filing

2. Executive Summary

Automated network analysis identified a cluster of 5 accounts exchanging funds in a pattern consistent with layering: money moved through 3 intermediary hops within a 4-day window, with two same-day transactions structured just under the CAD \$10,000 reporting threshold. The cluster moved a total of approximately \$87,450 CAD, exceeding this project's \$50,000 CAD escalation threshold for STR-recommended cases. The network was surfaced by the dashboard's community-detection layer after an initial transaction (8006345A0 → 800FC2A60, \$19,496.68) was flagged by the anomaly-scoring model.

3. Subject Accounts

Account ID	Associated Name (synthetic)	Role in Network	Connected Txns
8006345A0	Fictional Holdings Ltd. (synthetic entity)	Originator	3
800FC2A60	J. Whitfield (synthetic individual)	Intermediate	4
80357D960	Northbend Trading Co. (synthetic entity)	Intermediate	2
80357E110	M. Corrigan (synthetic individual)	Terminal / cash-out	3

8001C97D0	R. Alaoui (synthetic individual)	Terminal / cash-out	2
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4. Detection Methodology

- Stage 1 (transaction-level): each transaction was scored for anomaly likelihood using an unsupervised Isolation Forest model, with a supervised XGBoost classifier run in parallel for comparison and model-monitoring purposes.
- Stage 2 (network-level): accounts connected to any flagged transaction were extracted into a subgraph; Louvain community detection grouped the subgraph into clusters ("rings") of tightly interconnected accounts.
- Escalation rule: any detected ring moving more than CAD \$50,000 in intra-cluster transactions is flagged for STRstyle investigation write-up, consistent with this project's Strategy Document.

5. Transaction Timeline

Date	From Account	To Account	Amount (CAD)	Type	LaunderingLabeled
2026-01-03	8006345A0	800FC2A60	\$19,496.68	Transfer	Yes
2026-01-03	800FC2A60	80357D960	\$18,900.00	Transfer	Yes
2026-01-04	80357D960	80357E110	\$9,450.00	Cash-out	Yes (near \$10K threshold)
2026-01-04	80357D960	80357E110	\$9,200.00	Cash-out	Yes (near \$10K threshold)
2026-01-05	800FC2A60	8001C97D0	\$17,800.00	Transfer	No
2026-01-06	8001C97D0	80357E110	\$12,600.00	Transfer	No

6. FINTRAC Indicators Matched

The following indicators, drawn from FINTRAC's published money-laundering typologies, are present in this case:

FINTRAC Indicator	Basis in This Case
Structuring	Two same-day transactions of \$9,450 and \$9,200 between the same account pair, both just under the CAD \$10,000 reporting threshold.
Rapid movement of funds	Funds moved through 3 intermediary accounts within a 4-day window, consistent with a layering pattern rather than ordinary account activity.
Unusual network topology	Community detection (Louvain algorithm) grouped these 5 accounts into a single cluster with materially higher intra-cluster transaction density than the surrounding network.
Round-number / threshold-adjacent amounts	Multiple transaction amounts cluster just below the \$10,000 CAD threshold rather than showing the smoother distribution expected of ordinary commercial activity.

7. Grounds for Suspicion

The combination of (a) two same-day transactions immediately below the CAD \$10,000 threshold between the same account pair, (b) rapid multi-hop fund movement inconsistent with the accounts' prior transaction history, and (c) a network topology isolated from the broader transaction graph, together meet this project's criteria for recommending further investigation. No single indicator alone would have triggered escalation; it is the combination, surfaced only by combining transaction-level scoring with network-level analysis, that supports the recommendation below.

8. Recommended Action

- Escalate to AML/Compliance for manual review of accounts 8006345A0, 800FC2A60, 80357D960, 80357E110, and 8001C97D0.

- File a Suspicious Transaction Report with FINTRAC if manual review confirms the pattern (illustrative recommendation for this portfolio case study).
- Apply enhanced monitoring to all 5 accounts for 90 days following escalation.

9. Methodology & Data Disclosure

This investigation summary was generated as part of a portfolio project combining real Canadian fraud-report data (Canadian Anti-Fraud Centre) with a synthetic transaction-scoring and network-detection engine (PaySim and the IBM AML dataset). Synthetic data was used for the transaction- and network-level analysis because real bank transaction data cannot legally be obtained for a project of this kind under PIPEDA and the Bank Act. This approach mirrors standard practice in the fraud/AML analytics field, where synthetic datasets such as PaySim and IBM's AMLSim are widely used for model prototyping and demonstration.